PORTFOLIO: MULTI-OMICS TARGET DISCOVERY IN PANCREATIC CANCER

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1. **Disease and Question**

   Pancreatic ductal adenocarcinoma (PDAC) has a five-year relative survival of about 13% (Siegel et al., 2025) and few effective targeted treatments despite well-characterized driver genes. This package combines completed multi-omics integration, sequence-based druggability triage, and structural assessment to identify candidates for experimental validation. A good target in this context is increased in tumor at both RNA and protein, has a plausible therapeutic modality, and, for a small molecule, a confident binding pocket (criteria C1-C3, section 5). XDH was prioritized first, MUC16 second, and CEL was retained conditionally. These are just research priorities.

   CPTAC-PDAC data were obtained through LinkedOmicsKB (Cao et al., 2021; Liao et al., 2023). RNA and protein tumor versus normal summaries were combined with somatic mutation frequency using version-stripped Ensembl gene IDs. Patient pairing was not used in the group comparison, therefore, integration was performed at the gene level.

2. **Multi-Omics Integration**

   The three way join retained 1880 genes. Protein and mutation coverage were the main restriction, 1882 of 3362 gene in the mutation table were present in proteomics, and 1880 also had RNA data. RNA and protein changes had matching signs for 1190 genes, where 690 were discordant. The lab 2 score was the equal-weighted mean of rank percentiles for absolute RNA change, absolute protein change, and mutation frequency.

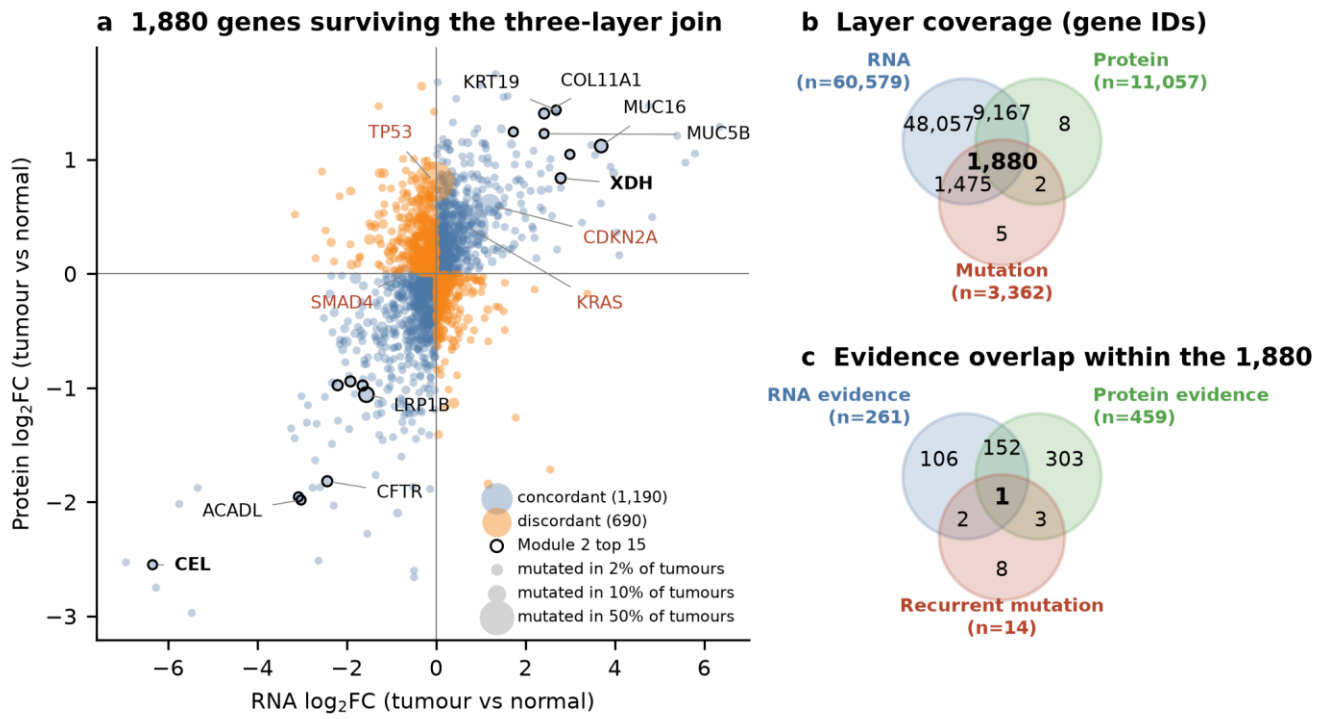


Figure 1. Multi-omics integration. **(a)** RNA versus protein log2 change for 1,880 genes (blue indicates matching signs, orange opposite signs, marker size mutation frequency, and black rings the Lab 2 top 15). **(b)** Gene availability across layers. **(c)** Selected evidence within the intersection: RNA |log2FC| ≥1 and protein |log2FC| ≥0.5, each at BH FDR <0.05, plus mutation frequency ≥5%. FDR was calculated across the full source table for each expression layer. MUC16 alone meets all three thresholds.

Multi-omics interpretation

No primary analysis was rerun, Benjamini-Hochberg adjustment and explicit thresholds were added to summarize the existing outputs. After full-layer adjustment, 261 retained genes met the RNA threshold, 459 met the protein threshold and 14 met the mutation threshold. Of the 153 genes meeting both expression thresholds, 140 were concordant. Within the 1880-gene intersection, MUC16 was the only gene meeting all three selected thresholds.

Mutation frequency provided limited separation for many genes, approximately 82% of retained genes had a frequency of 0.95%. Many mutation percentiles were therefore tied, leaving expression differences to distinguish those genes. LRP1B had a relatively high mutation frequency (8.6%), but its protein change was not significant after full-layer adjustment (FDR 0.45). A high integrated rank should not be read as strong support in every layer.

Known biology was recovered through MUC16 (rank1), COL11A1 (3), and KRT19 (10), representing tumor-associated, stromal, and ductal signs. Major drivers ranked lower are CDKN2A 32, KRAS 69, TP53 381, and SMAD4 530. KRAS was mutated in 96% of tumors but

showed modest abundance changes, illustrating how equal weighting can dilute mutation-driven biology. RNF43 was lost because the protein data was absent.

TP53 provides a useful discordant example, RNA change was nearly zero (-0.046), protein increased (+0.803), and mutation frequency was 82.9%. Altered turnover or mutant stabilization could explain protein accumulation without increased transcription, but neither mechanism was established by these data. XDH showed concordant increases (RNA +2.788; protein +0.836; FDR $<10^{-6}$ in both layers).

3. **Druggability triage**

All 15 Lab 2 candidates were mapped to ESM C embedding. Logistic-regression priors were trained on 1201 proteins with clinical-status labels, and none of the selected candidates overlapped the training panel. The embeddings provide sequence-based representations, rather than direct measurements of binding pockets (EvolutionaryScale, 2024).

Family-held-out validation gave AUROC 0.732 and average precision 0.548, compared with 0.848 and 0.727 under random splitting. The positive-label prevalence was 0.334. Holding out protein families reduces overoptimism caused by closely related proteins appearing on both sides of a split. These metrics were measured for any clinical label, not separately for the two modality models.

Lab 3 combined the within-shortlist percentile of the Lab 2 score (60%) with the percentile of the highest modality prior (40%). Greater weight was retained for disease evidence because the prior is not PDAC-specific. The weak observed association between disease score and tractability (Spearman $\rho = -0.14$; $n = 15$) suggests different information, but does not establish statistical independence.

Sequence triage and structural assessment

Six candidates had a modality prior ≥0.4, and they are CEL, XDH, and FN1 favored small molecules, MUC5B, NPTX2, and LRP1B favored antibodies. CEL led the Lab 3 ranking (p_{sm} 0.716), whereas XDH ranked sixth (p_{sm} 0.503). CFTR scored poorly despite established druggability, showing that model outputs should be checked against external evidence. MUC16 also scored low ($p_{a\beta}$ 0.151), although an anti-MUC16 ADC was investigated in a phase I study including pancreatic cancer (Liu et al., 2016). This supports modality precedent, not proven PDAC benefit. Sequence truncation is a possible explanation for its low score, not a verified cause.

Eight of the 15 candidates decreased at both RNA and protein levels, including CEL, CFTR, LRP1B, and NPTX2. Such changes can reflect tissue composition or altered cell states. Increased expression supports investigating inhibition, but does not demonstrate tumor dependency, and decreased expression leaves therapeutic direction unresolved.

4. Structural assessment

CEL and XDH were selected for structural assessment because they had the strongest small-molecule priors. Existing AlphaFold models were compared with experimental structures, and cavities were assessed with fpocket (Jumper et al., 2021; AlphaFold Protein Structure Database, n.d.; Le Guilloux et al., 2009). Pockets were rejected when mean lining-residue pLDDT was <70 and flagged when internal PAE exceeded 10 Å.

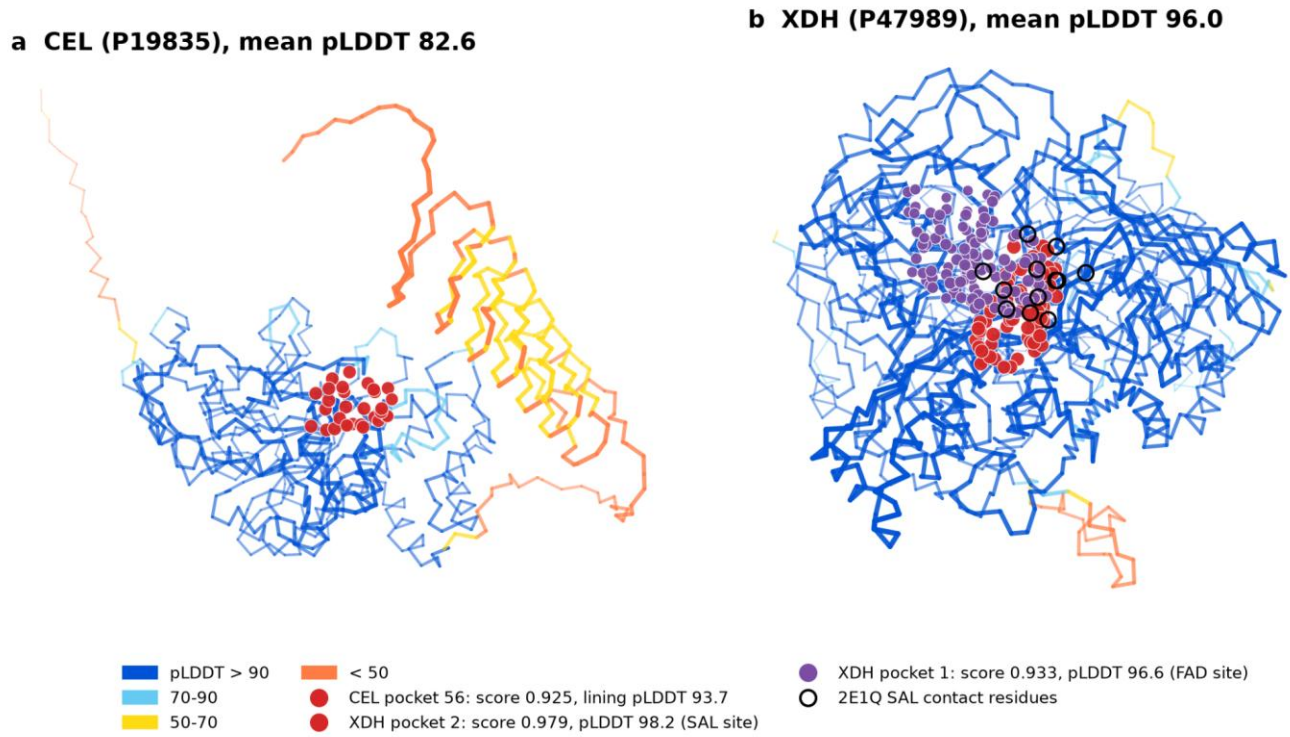


Figure 2. Confidence-colored AlphaFold models. Dark blue: pLDDT >90; cyan: 70–90; yellow: 50–70; orange: <50. Spheres show fpocket lining atoms. CEL pocket 56 and XDH pocket 2 are red; XDH pocket 1 is purple. Black rings identify the saved SAL-contact residues mapped from 2E1Q chain A. The colored atoms represent pocket lining, not a measured ligand pose.

Table 1. Structural evidence carried forward from Lab 4.

Target	Global confidence	Best retained pocket	Experimental comparison
CEL	Mean pLDDT 82.6; 66.3% >90	56; score 0.925; lining pLDDT 93.7; PAE 2.1 Å	1F6W A: RMSD 1.35 Å over 527 confident identical pairs
XDH	Mean pLDDT 96.0; 93.4% >90	2; score 0.979; lining pLDDT 98.2; PAE 1.5 Å	2E1Q A: RMSD 0.48 Å over 1,297 confident identical pairs

Structural interpretation and final prioritization

XDH pocket 2 overlapped 92% of mapped SAL-contact residues, and pocket 1 overlapped 90% of FAD contacts. These results support a plausible binding environment, but contact overlap does not establish ligand pose, affinity, or selectivity. The 2E1Q reference contains E803V, including a SAL-contact position, so transfer to wild-type XDH remains uncertain (Yamaguchi et al., 2007). Residues 168-190 were low confidence, consistent with flexibility in a conversation-sensitive region, rather than proof of a particular conformation.

CEL had a confident central group at residues 21-552, with low-confidence stretches at 3-17, 555-592, and 717-753. Pocket 56 passed the confident filters, but lacked experimental ligand-site validation. Pocket 60 was rejected despite a score of 0.562 because its mean pLDDT was 51.1 and PAE 13.4 Å. Agreement with 1F6W supports the aligned catalytic region, not the full-length tail or activated pocket geometry (Terzyan et al., 2000).

5. **Prioritized target package**

Three criteria guided the handoff, with C1, increased RNA and protein, with C2, a best modality prior ≥0.4 or established clinical investigation, with C3, a retained pocket with acceptable pLDDT/PAE and fpocket score ≥0.5. The 0.4 and 0.5 cutoffs are analyst-selected triage thresholds, not universal druggability standards. Final ordering also considers evidence gaps and target-specific biology.

	Module 2 rank	RNA log_2FC	Protein log_2FC	Mutated (%)	Module 3 rank	p_{sm}	p_{ab}	Structure (Module 4)	Package call
1. XDH	9	+2.79	+0.84	2.9	6	0.50	0.00	pocket 0.979	Advance
2. MUC16	1	+3.70	+1.12	5.7	2	0.00	0.15	not assessed	Advance
3. CEL	4	-6.35	-2.55	1.9	1	0.72	0.01	pocket 0.925	Conditional
MUC5B	13	+2.42	+1.23	1.9	11	0.05	0.55	not assessed	Watch
FN1	15	+1.73	+1.24	1.9	12	0.48	0.29	not assessed	Watch
COL11A1	3	+2.42	+1.40	2.9	4	0.10	0.00	not assessed	Low tractability
KRT19	10	+2.69	+1.43	1.9	13	0.01	0.00	not assessed	Low tractability
EPS8L3	14	+2.99	+1.04	1.9	15	0.09	0.00	not assessed	Low tractability
LRP1B	5	-1.56	-1.06	8.6	3	0.28	0.47	not assessed	Down in tumour
BRSK2	6	-3.08	-1.96	1.9	5	0.27	0.00	not assessed	Down in tumour
CFTR	2	-2.44	-1.82	2.9	7	0.04	0.00	not assessed	Down in tumour
ACADL	7	-3.03	-1.98	1.9	8	0.15	0.00	not assessed	Down in tumour
NOS1	8	-2.21	-0.98	2.9	9	0.21	0.00	not assessed	Down in tumour
NPTX2	11	-1.92	-0.94	2.9	10	0.05	0.50	not assessed	Down in tumour
LAMC3	12	-1.65	-0.98	2.9	14	0.10	0.08	not assessed	Down in tumour

Figure 3. Evidence matrix for all 15 candidates. Values and Lab 2–3 ranks are carried forward from saved outputs. “Advance” means advance to experimental validation. “Low tractability” indicates limited support from this model and reviewed evidence, not proven inability to drug the protein. Missing structures are evidence gaps. The shortlist is an

analyst-defined interpretation; the Lab 3 numerical ranking is preserved. “Module 2” and “Module 3” in the column headers refer to Labs 2 and 3.

Prioritized shortlist and Validation

Table 2. Final shortlist and the next validation decision

Priority	Candidate and rationale	Next decision
1 XDH	Increased RNA/protein, small-molecule prior 0.503, and a confident pocket overlapping experimental ligand contacts. Approved inhibitors (allopurinol and febuxostat; Pacher et al., 2006) provide a repurposing test.	Establish the expressing compartment and test whether PDAC cells depend on XDH.
2 MUC16	Lab 2 rank 1 and prior ADC investigation support follow-up. Structure and PDAC-specific benefit remain unestablished.	Confirm accessible surface expression and evaluate an appropriate antibody/ADC approach.
3 CEL	Lab 3 rank 1 and a retained pocket support tractability, but strong depletion leaves therapeutic direction unresolved.	Hold until cellular origin and functional direction are resolved.

XDH was placed first because disease direction, sequence-based tractability, and structural support provide complementary evidence. MUC16 were placed above CEL because its unresolved structural assessment is a data gap, while CEL also has an unresolved intervention direction. MUC16 exceeded the 2700 residue limit of the lab workflow, this does not exclude domain-level modelling. MUC5B and FN1 remain on a watch list, because secreted MUC5B could create an antigen sink, while FN1 precedent concerns the EDB-containing isoform, which these gene level data cannot resolve.

6. Limitations and validation

The comparisons describe gene-level group differences without using patient pairing. Bulk tumor composition, unequal normal-group sizes, and incomplete layer coverage can affect the ranking. Absolute expression changes do not establish therapeutic direction, and mutation frequency does not distinguish drivers from passengers. The top 15 scores are tightly spaced (0.932-0.979), making exact ranks sensitive to weighting. Sequence priors are moderately predictive, and only two candidates received structural assessments. Predicted pockets omit the important biological context, including glycans, cofactors, and conformational changes.

Validation should first localize XDH expression using a suitable experiment such as single-cell data, followed by protein and enzyme-activity measurements in PDAC and nonmalignant pancreatic models. Multiple independent genetic perturbations and negative controls would

test dependency. Drug dose responses should be compared with genetic effects and target engagement to assess specificity and normal-cell toxicity. Binding and activity assays using appropriately cofactor-loaded wild-type protein would address the mutant-reference uncertainty.

For CEL, association with acinar markers such as PRSS1 and CPA1 would support a tissue-composition explanation but would not settle therapeutic direction. Functional reduction and restoration experiments are needed. MUC16 follow-up should establish surface accessibility in PDAC and assess relevant membrane-proximal domains. None of the current results demonstrate treatment efficacy.

Analysis files

Lab 2–4 notebooks, targets_pdac.csv, targets_pdac_w3.csv, structure_assessment_pdac.csv, and portfolio figures:
https://github.com/Deepthi-Dineshkumar/biot6900/tree/main
Primary results were carried forward from the completed lab outputs.

Reference

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